

MOST COMMONLY ASKED INTERVIEW QUESTIONS

Theory Based Questions

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2. What is Power BI and what are its key components?
3. Explain the difference between Power BI Desktop, Power BI Service, and Power BI Mobile.
4. What are the advantages of using Power BI over other BI tools?
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1. What is Business Intelligence (BI) and why is it important?

Business Intelligence (BI) refers to the technologies, processes, and tools used to collect, integrate, analyze, and present business data.

It helps organizations make data-driven decisions by providing actionable insights.

BI combines historical data, current operations, and predictive analytics.

It involves reporting, dashboards, data visualization, and trend analysis.

The main goal is to improve efficiency, spot opportunities, and reduce risks.

2. What is Power BI and what are its key components?

Power BI is Microsoft's Business Intelligence tool for data analysis and visualization.

It helps users connect to multiple data sources, transform data, and create interactive reports.

Key components include Power BI Desktop, Power BI Service, and Power BI Mobile.

It supports features like DAX, Power Query, and custom visuals.

Power BI is widely used for making business insights accessible to both technical and non-technical users.

3. Explain the difference between Power BI Desktop, Power BI Service, and Power BI Mobile.

Power BI Desktop is a Windows application used to create and design reports.

Power BI Service is the cloud-based platform for publishing, sharing, and collaborating on reports.

Power BI Mobile is an app for viewing and interacting with reports on mobile devices.

Desktop is mainly for development, Service for sharing, and Mobile for consumption.

Together, they form the complete Power BI ecosystem.

4. What are the advantages of using Power BI over other BI tools?

Power BI offers a user-friendly drag-and-drop interface with strong integration to Microsoft products.

It supports a wide range of data sources with both Import and DirectQuery modes.

It is cost-effective compared to many BI tools, with a free version for beginners.

It has powerful data modeling features like DAX and Power Query.

Regular updates from Microsoft ensure constant improvements and new features.

5. What is the Power BI workflow?

The Power BI workflow starts with connecting to data sources.
Next, the data is cleaned, transformed, and modeled in Power BI Desktop.
Reports are designed with visuals and interactivity features.
These reports are then published to the Power BI Service for sharing.
Users can view and interact with reports on web browsers or mobile apps.

6. What is the role of a Quadrant report in BI analysis?

A Quadrant report divides data into four sections for comparative analysis.
It is often used to compare performance against two key metrics.
For example, it can help identify top performers, underperformers, and average performers.
This visualization makes spotting patterns and anomalies easier.
It is widely used in market analysis, product performance, and competitive analysis.

7. Name a few competitors of Power BI and how it differs from them.

Competitors include Tableau, QlikView, Looker, and Google Data Studio.
Power BI is generally more affordable and integrates better with Microsoft tools.
Tableau is known for more advanced visuals, but Power BI is easier for beginners.
QlikView offers strong data processing speed, but Power BI is more intuitive.
Power BI has faster adoption due to its Microsoft ecosystem and low entry cost.

8. What are the main building blocks of Power BI?

The main building blocks are datasets, reports, dashboards, and tiles.
Datasets are collections of data from various sources.
Reports are collections of visuals created from datasets.
Dashboards are single-page summaries combining visuals from multiple reports.
Tiles are the individual visuals or charts within a dashboard.

9. What is the difference between a dashboard and a report in Power BI?

A report is a multi-page collection of visuals, tables, and charts.

A dashboard is a single page with selected visuals from one or more reports.

Reports allow detailed exploration and filtering, while dashboards provide a quick overview.

Dashboards are available only in Power BI Service, not Desktop.

Reports are for deep analysis, dashboards for high-level monitoring.

10. What are some limitations of Power BI?

Power BI has dataset size limits depending on the license type.

Complex DAX calculations can affect performance.

Free users have limited sharing options compared to Pro or Premium users.

DirectQuery mode may have slower performance than Import mode.

Some advanced features require additional licensing costs.

11. What are the different data import options available in Power BI?

Power BI supports importing data from files like Excel, CSV, XML, and JSON.

It can connect to databases such as SQL Server, MySQL, and Oracle.

It supports cloud services like Azure, Google Analytics, and SharePoint.

You can also connect to APIs and web data sources.

Connections can be in *Import* or *DirectQuery* mode.

12. What is the difference between "Import" and "DirectQuery" modes?

Import mode loads a copy of the data into Power BI for faster performance.

DirectQuery keeps data in the source and queries it live when needed.

Import mode allows more transformations and modeling flexibility.

DirectQuery ensures real-time data updates but may be slower.

Import mode is better for smaller datasets, DirectQuery for large, live datasets.

13. What is Power Query in Power BI?

Power Query is the data preparation and transformation tool in Power BI. It allows cleaning, shaping, and combining data before analysis. It uses a scripting language called M for transformations. It is accessible in Power BI Desktop through the Power Query Editor. Its main goal is to prepare raw data into a usable format for reporting.

14. What are the benefits of data transformation before visualization?

It ensures data accuracy by removing errors and inconsistencies. It improves performance by cleaning unnecessary columns and rows. It makes data more understandable with proper formatting. It helps combine multiple data sources into a single model. Clean data leads to more reliable insights and better decision-making.

15. What is the purpose of the Queries pane in Power Query Editor?

The Queries pane shows all loaded and created queries in Power BI. It allows switching between different datasets during transformation. You can rename, duplicate, or delete queries from this pane. It also organizes queries into groups for better management. This helps in handling multiple data sources within one report.

16. What is M Language in Power BI?

M Language is the formula language used in Power Query. It defines the steps for data extraction and transformation. Every change in Power Query generates M code in the background. It is case-sensitive and follows functional programming concepts. Advanced users can write M code manually for complex transformations.

17. What are the main data types supported in Power BI?

Power BI supports data types like Whole Number, Decimal Number, and Percentage. It also supports Date/Time, Date, and Time types for temporal data. Text and Boolean (True/False) types are available for categorical data. Binary type is used for storing images and files. Choosing the right type ensures correct calculations and filtering.

18. How can you change the data type of a column in Power BI?

Open the dataset in Power Query Editor. Select the column whose data type you want to change. Choose the correct type from the data type dropdown in the toolbar. You can also right-click the column and select "Change Type". Changing the type affects how the column is used in calculations.

19. What is the difference between inbuilt column transformations and row transformations in Power Query?

Column transformations modify entire columns, such as splitting, merging, or renaming. They can also change data types, apply calculations, or format text. Row transformations affect rows, such as filtering, sorting, or removing duplicates. Both are used to clean and prepare data for analysis. Column transformations change structure, row transformations change data subset.

20. Explain the difference between “Append Queries” and “Merge Queries”.

Append Queries combine rows from two or more tables with the same structure. It is similar to stacking datasets vertically. Merge Queries combines columns from related tables based on a matching key. It is similar to joining tables in SQL. Append adds more data, Merge adds more details.

21. What is Power Pivot in Power BI?

Power Pivot is a data modeling component in Power BI.
It allows creating relationships between multiple tables.
It supports calculated columns, measures, and advanced DAX formulas.
It enables building complex data models for analysis.
Originally part of Excel, it is now fully integrated into Power BI.

22. What is data modeling in Power BI?

Data modeling is the process of structuring and relating datasets.
It defines relationships between tables for better analysis.
It involves creating calculated columns, measures, and hierarchies.
A well-designed model improves performance and accuracy.
It is the backbone of interactive and reliable reports.

23. What is a relationship in Power BI?

A relationship connects columns from two tables to link data.
It enables combined analysis from different datasets.
Relationships can be one-to-one, one-to-many, or many-to-many.
They are created using matching fields, like IDs or keys.
Without relationships, cross-table filtering and calculations won't work properly.

24. What is the difference between cardinality types (One-to-One, One-to-Many, Many-to-Many)?

One-to-One means each value in one table matches exactly one in another.
One-to-Many means one value in a table matches multiple in another.
Many-to-Many means multiple values in both tables can match each other.
Choosing the wrong cardinality can cause incorrect results.
Cardinality defines how tables interact in the model.

25. What is the difference between calculated columns and measures in Power BI?

Calculated columns are created row-by-row in a table.

Measures are calculations performed on the fly, based on filters.

Calculated columns consume more memory as they are stored in the model.

Measures are more efficient for aggregations and summaries.

Use columns for fixed data, measures for dynamic calculations.

26. What is DAX in Power BI?

DAX stands for Data Analysis Expressions.

It is a formula language used for calculations in Power BI.

DAX is used in calculated columns, measures, and calculated tables.

It includes functions for math, text, logic, time intelligence, and filtering.

It works with data models to produce meaningful insights.

27. Write the basic syntax structure of a DAX formula.

A DAX formula generally follows this structure:

`<NewColumnName> = <FunctionName>(<Arguments>)` for columns.

For measures: `<MeasureName> = <FunctionName>(<Arguments>)`.

It always starts with the result name followed by an equal sign.

Functions can be nested and combined for complex logic.

28. What is the difference between calculated tables and calculated columns?

Calculated tables are new tables created from existing data using DAX.

Calculated columns are added fields within existing tables.

Tables can be used for creating lookup data or summaries.

Columns store row-level calculations.

Tables are model-level objects, columns are table-level objects.

29. What is a Date Dimension table and why is it important?

A Date Dimension table contains continuous date values.
It is used for time-based analysis like year-to-date or month-to-date.
It improves performance in time intelligence functions.
It allows creating custom date hierarchies.
It is essential for accurate time calculations in DAX.

30. What is the RELATED function used for in Power BI?

The RELATED function fetches values from another table using an existing relationship.
It works in calculated columns within the related table.
It's useful for bringing lookup values into the current table.
It only works when a valid relationship exists.
It simplifies combining data without using Merge Queries.

31. What are DAX logical functions? Give examples.

Logical functions return TRUE or FALSE based on conditions.
Examples include IF, AND, OR, NOT, and SWITCH.
They are used for conditional calculations in measures and columns.
They help in creating dynamic logic based on filters.
For example, IF(Sales > 5000, "High", "Low").

32. What are DAX text functions?

Text functions manipulate and format string data.
Examples include CONCATENATE, LEFT, RIGHT, MID, and SEARCH.
They help in creating labels, extracting substrings, or formatting text.
They are useful for building descriptive fields in reports.
For example, joining first and last name into a full name.

33. What are DAX mathematical and statistical functions?

These functions perform numeric calculations and statistical analysis.
Examples include SUM, AVERAGE, ROUND, MAX, MIN, and STDEV.
They help in aggregating data, rounding values, and measuring variability.
Statistical functions are used for variance, standard deviation, etc.
They are core to financial, sales, and operational analysis.

34. What are DAX aggregation functions?

Aggregation functions summarize values from a column or table.
Examples include SUM, COUNT, AVERAGE, MAX, and MIN.
They are used to produce KPIs and totals.
They work dynamically with report filters.
They are essential for summarizing large datasets.

35. What are DAX filter functions?

Filter functions return a subset of data based on conditions.
Examples include FILTER, ALL, ALLEXCEPT, and CALCULATE.
They control the data context for calculations.
They are used to modify or override filters in visuals.
They are powerful for complex business logic.

36. What are DAX time intelligence functions?

These functions perform calculations over time periods.
Examples include DATEADD, SAMEPERIODLASTYEAR, TOTALYTD.
They help in year-to-date, month-to-date, and growth comparisons.
They require a Date Dimension table for accuracy.
They are essential for trend and historical analysis.

37. What is the difference between CALCULATE and CALCULATETABLE in DAX?

CALCULATE returns a single value after applying filters.

CALCULATETABLE returns an entire table after filtering.

Both change the filter context of a calculation.

Use CALCULATE for measures and aggregations.

Use CALCULATETABLE when a table result is required.

38. What is the ALL function used for in DAX?

The ALL function removes filters from a table or column.

It is used to calculate values ignoring current filter context.

It is common in percentage or rank calculations.

It helps in comparing filtered values to overall totals.

For example, market share calculations.

39. What is the difference between FILTER and ALL in DAX?

FILTER returns a table with rows meeting a specific condition.

ALL returns all rows, ignoring filters applied in visuals.

FILTER narrows down results, ALL expands them.

They can be combined for advanced logic.

FILTER is condition-based, ALL is context-resetting.

40. What is the purpose of the EARLIER function in DAX?

EARLIER retrieves the value of a column from an earlier row context.

It is often used in calculated columns for row-by-row comparisons.

Commonly used in running totals or rankings.

It works with nested row contexts created by iterators.

It is replaced by variables in many modern DAX scenarios.

41. What are the different visualization types in Power BI?

Power BI supports bar, column, line, pie, donut, and area charts. It also includes scatter plots, maps, tables, and matrices. Special visuals include KPIs, cards, and gauges. Users can also import custom visuals from the marketplace. Different visuals are used based on data type and purpose.

42. What is a slicer in Power BI?

A slicer is a visual filter used to select data in a report. It allows users to interactively filter other visuals. It supports single or multiple selections. Common slicers include date, category, and numeric range. They make reports more dynamic and user-friendly.

43. What is the difference between slicers and filters?

Slicers are visual elements displayed on the report canvas. Filters can be applied at visual, page, or report level in the filter pane. Slicers are more interactive for end-users. Filters are more hidden and structured. Both achieve similar outcomes but with different user experiences.

44. What are KPI visuals used for?

KPI visuals display key performance indicators. They show progress toward a target using visual cues. They typically include an actual value, target value, and trend. They help in quick business performance tracking. Commonly used for metrics like sales goals and efficiency rates.

45. What is the difference between a table and a matrix visual in Power BI?

A table displays data in rows and columns without hierarchy.
A matrix allows grouping, drilldown, and pivot-like summaries.
Matrices can show subtotals and grand totals.
Tables are best for simple data display.
Matrices are better for grouped or comparative analysis.

46. What is a drillthrough in Power BI?

Drillthrough lets users navigate to a detailed page filtered for a specific value.
It is triggered by right-clicking a data point.
It creates a focused report page for deeper analysis.
It is useful for breaking down high-level KPIs.
It keeps reports clean while offering detail when needed.

47. What is the difference between drillthrough and drilldown?

Drilldown explores data hierarchy within the same visual.
Drillthrough navigates to another page with filtered data.
Drilldown works on fields like Year → Month → Day.
Drillthrough works based on a selected data point.
Drilldown is intra-visual, drillthrough is cross-page.

48. What is conditional formatting in Power BI?

Conditional formatting changes visual appearance based on rules.
It can be applied to colors, backgrounds, font size, and data bars.
It highlights important trends or exceptions.
It improves readability and draws attention to key values.
It can be rule-based or dynamic using DAX measures.

49. What are visual interactions in Power BI?

Visual interactions define how one visual affects another on the same page.
They can be set to filter, highlight, or do nothing.
They control the flow of data between visuals.
They are configured in "Edit Interactions" mode.
They enhance user control over report analysis.

50. What is the purpose of creating buttons in Power BI reports?

Buttons are interactive elements for navigation or triggering actions.
They can link to pages, bookmarks, or external URLs.
They make reports more user-friendly and interactive.
Buttons can also be used for showing/hiding visuals.
They enhance dashboard navigation and design.

51. What is Power BI Service?

Power BI Service is Microsoft's cloud-based platform for hosting and sharing reports.
It allows collaboration, dashboard creation, and real-time data updates.
Users can view, interact, and share reports through a browser.
It supports scheduled refresh, RLS, and app workspaces.
It connects Power BI Desktop reports to an online environment.

52. What is the difference between a report and a dashboard in Power BI Service?

A report is a multi-page collection of visuals created in Power BI Desktop.
A dashboard is a single-page view of pinned visuals from one or more reports.
Reports allow deep analysis, dashboards provide quick insights.
Dashboards are only available in Power BI Service.
Reports are detailed; dashboards are summaries.

53. What is the difference between Publish to Web and Share in Power BI?

Publish to Web generates a public link for anyone to view the report.
Share sends access only to specific licensed users.
Publish to Web has no security and is for public data.
Share is secure and used for private company data.
Both allow report access, but with different privacy levels.

54. What are the steps to create a dashboard in Power BI Service?

Publish your Power BI Desktop report to the Service.
Open the report in Power BI Service.
Pin visuals from the report to a new or existing dashboard.
Arrange and resize visuals as needed.
Save and share the dashboard with others.

55. What is Row-Level Security (RLS) in Power BI?

RLS restricts data access based on user roles.
It ensures users only see the data relevant to them.
It is implemented using DAX filter rules.
RLS can be static (predefined roles) or dynamic (based on user login).
It enhances security in shared reports.

56. What is the difference between Static RLS and Dynamic RLS?

Static RLS has fixed filter rules for each role.
Dynamic RLS filters data based on the logged-in user.
Static is easier to set up but less flexible.
Dynamic uses a lookup table with user IDs or emails.
Dynamic is better for large organizations with many users.

57. What are the ways to share a Power BI report?

Share directly with users in Power BI Service.
Publish to Web for public access.
Export as PDF or PowerPoint.
Embed in SharePoint, Teams, or websites.
Share as an app from a workspace.

58. What are the limitations of publishing to web in Power BI?

Reports become publicly accessible with no security.
Sensitive or confidential data should not be used.
Some visuals and features may not be supported.
Data refresh frequency depends on the Service plan.
Published links can be indexed by search engines.

59. What is the difference between a workspace and an app in Power BI Service?

A workspace is a collaborative area where reports and datasets are built.
An app is a packaged, read-only version of reports and dashboards from a workspace.
Workspaces are for creators; apps are for end-users.
Apps provide a cleaner, user-friendly interface.
Only Pro or Premium users can publish apps.

60. What are some best practices for designing Power BI dashboards?

Keep the design clean and focused on key metrics.
Use consistent colors, fonts, and formatting.
Limit visuals per page for better performance.
Use slicers and filters for interactivity.
Test for mobile view and different screen sizes.

61. What is the difference between Star Schema and Snowflake Schema in Power BI?

Star Schema

1. Central fact table connected directly to dimension tables.
2. Simple and easy to understand.
3. Faster query performance (fewer joins).
4. Uses slightly more storage due to repeated data.
5. Preferred in Power BI for efficiency and best practices.

Snowflake Schema

1. Fact table connected to dimensions, but dimensions are split into sub-tables (normalized).
2. Complex and harder to understand.
3. Slower query performance (more joins).
4. Uses less storage by avoiding redundancy.
5. Not recommended in Power BI, unless normalization is required.

62. What is the difference between M Language and DAX in Power BI?

M Language (Power Query Formula Language)

1. Used in Power Query for data transformation (loading, cleaning, shaping).
2. Works before data is loaded into the data model.
3. Syntax looks like functional programming (similar to F#).
4. Example tasks: Remove duplicates, merge tables, change data types.
5. Runs in the Query Editor stage.

DAX (Data Analysis Expressions)

1. Used in Power BI data model for calculations and analysis.
2. Works after data is loaded into the data model.
3. Syntax is like Excel formulas.
4. Example tasks: Create measures, calculated columns, time intelligence functions.
5. Runs in the Data Model / Report View stage.